Week 1

Server Output

```
PS C:\Users\Lenovo\networkprogramming2025\week01-tcp-client-server-basic> python server.py
[SERVER] Persistent server listening on 127.0.0.1:5000
[SERVER] Press Ctrl+C to stop the server.
[SERVER] Connection from ('127.0.0.1', 53934)
[SERVER] Received empty message
[SERVER] Closed connection with ('127.0.0.1', 53934)
[SERVER] Connection from ('127.0.0.1', 53935)
[SERVER] Received: Hello Wiseza
[SERVER] Closed connection with ('127.0.0.1', 53935)
```

Client Output

```
PS C:\Users\Lenovo\networkprogramming2025\week01-tcp-client-server-basic> python client.py
[CLIENT] Sending: Hello Wiseza
[CLIENT] Received: ACK: Hello Wiseza
PS C:\Users\Lenovo\networkprogramming2025\week01-tcp-client-server-basic>
```

Week 2

Receiver Output

```
PS C:\Users\Lenovo\networkprogramming2025\week02-udp-unicast-basic> python receiver.py
[RECEIVER] Listening on 127.0.0.1:6000
[RECEIVER] From ('127.0.0.1', 52273): เป็นยังไงบ้าง
[RECEIVER] From ('127.0.0.1', 63061): ฉันสบายดีสุดๆ
```

Sender Output

```
PS C:\Users\Lenovo\networkprogramming2025\week02-udp-unicast-basic> python sender.py
Enter message: เป็นยังไงบ้าง
[SENDER] Message sent
PS C:\Users\Lenovo\networkprogramming2025\week02-udp-unicast-basic> python sender.py
Enter message: ฉันสบายดีสุดๆ
[SENDER] Message sent
PS C:\Users\Lenovo\networkprogramming2025\week02-udp-unicast-basic>
```

Week 3

Broadcaster Output

```
PS C:\Users\Lenovo\networkprogramming2025\week03-udp-broadcast-basic> python broadcaster.py
Enter broadcast message: สวัสดีเพื่อนๆๆๆๆ
[BROADCASTER] Message sent
```

Listener Output (Terminal 1)

```
PS C:\Users\Lenovo\networkprogramming2025\week03-udp-broadcast-basic> python listener.py
[LISTENER] Listening for broadcast on port 7001
[LISTENER] From ('127.0.0.1', 61861): สวัสดีเพื่อนๆๆๆๆ
```

Listener Output (Terminal 2)

```
PS C:\Users\Lenovo\networkprogramming2025\week03-udp-broadcast-basic> python listener.py
[LISTENER] Listening for broadcast on port 7001
[LISTENER] From ('127.0.0.1', 61861): สวัสดีเพื่อนๆๆๆๆ
```

Week 4

Sender Output

```
PS C:\Users\Lenovo\networkprogramming2025\week04-udp-multicast-basic> python sender.py
Enter multicast message: เลิ้ปๆเลย
[SENDER] Multicast sent
```

Listener Output (Terminal 1)

```
PS C:\Users\Lenovo\networkprogramming2025\week04-udp-multicast-basic> python receiver.py
[RECEIVER] Joined 224.1.1.1:8000
[RECEIVER] ('192.168.1.44', 59806) -> เลิ้ปๆเลย
```

Listener Output (Terminal 2)

```
PS C:\Users\Lenovo\networkprogramming2025\week04-udp-multicast-basic> python receiver.py
[RECEIVER] Joined 224.1.1.1:8000
[RECEIVER] ('192.168.1.44', 59806) -> เลิ้ปๆเลย
```

Week 5

ฝั่ง Peer 1

```
PS C:\Users\Lenovo\networkprogramming2025\week05-peer-to-peer-basic> python peer.py 1
Send to peer ID:
[PEER 1] Listening on 9001

2
Message: เปิดเทอมเป็นยังไงบ้าง
[PEER 1] Sent to 2: เปิดเทอมเป็นยังไงบ้าง
Send to peer ID:
[PEER 1] From ('127.0.0.1', 53574): สบายดีเลยล่ะ
Send to peer ID:
```

ฝั่ง Peer 2

```
PS C:\Users\Lenovo\networkprogramming2025\week05-peer-to-peer-basic> python peer.py 2
Send to peer ID:
[PEER 2] Listening on 9002


[PEER 2] From ('127.0.0.1', 64122): เปิดเทอมเป็นยังไงบ้าง
Send to peer ID: 1
Message: สบายดีเลยล่ะ
[PEER 2] Sent to 1: สบายดีเลยล่ะ
Send to peer ID:
```

Week 6

Message Forwarding Works

- Messages from node 7000 สามารถส่งไปถึง node 7001 และ 7002 ได้

```
PS C:\Users\Lenovo\networkprogramming2025\week06-manet-basic> python node.py
[NODE 7000] Listening for neighbors...
```

Node 7001

```
PS C:\Users\Lenovo\networkprogramming2025\week06-manet-basic> python node.py
[NODE 7001] Listening for neighbors...
[NODE 7001] Received from ('127.0.0.1', 55703): Hello from node 7001 (TTL=3)
[NODE 7001] Peer 7002 unreachable
[NODE 7001] Peer 7002 unreachable
[NODE 7001] Received from ('127.0.0.1', 50316): Hello from node 7002 (TTL=3)
[NODE 7001] Received from ('127.0.0.1', 50318): Hello from node 7002 (TTL=2)
[NODE 7001] Received from ('127.0.0.1', 50344): Hello from node 7000 (TTL=3)
[NODE 7001] Received from ('127.0.0.1', 50346): Hello from node 7000 (TTL=2)
```

Node 7002

```
PS C:\Users\Lenovo\networkprogramming2025\week06-manet-basic> python node.py
[NODE 7002] Listening for neighbors...
[NODE 7002] Received from ('127.0.0.1', 50317): Hello from node 7002 (TTL=3)
[NODE 7002] Received from ('127.0.0.1', 50345): Hello from node 7000 (TTL=3)
[NODE 7002] Received from ('127.0.0.1', 50348): Hello from node 7000 (TTL=2)
[NODE 7002] Received from ('127.0.0.1', 50349): Hello from node 7000 (TTL=2)
[NODE 7002] Received from ('127.0.0.1', 50350): Hello from node 7000 (TTL=1)
```

Week 7

Terminal 1 → Node A

```
PS C:\Users\Lenovo\networkprogramming2025\week07-store-forward-basic> python node.py
[NODE 8000] Listening...
[NODE 8000] Peer 8001 unavailable, storing message
[NODE 8000] Peer 8002 unavailable, storing message
[NODE 8000] Received: Hello from node 8001 from ('127.0.0.1', 57271)
[NODE 8000] Sent stored message to 8001
[NODE 8000] Received: Hello from node 8002 from ('127.0.0.1', 57321)
[NODE 8000] Sent stored message to 8002
```

Terminal 2 → Node B

```
PS C:\Users\Lenovo\networkprogramming2025\week07-store-forward-basic> python node.py
[NODE 8001] Listening...
[NODE 8001] Peer 8002 unavailable, storing message
[NODE 8001] Received: Hello from node 8000 from ('127.0.0.1', 57296)
[NODE 8001] Received: Hello from node 8002 from ('127.0.0.1', 57322)
[NODE 8001] Sent stored message to 8002
```

Terminal 2 → Node C

```
PS C:\Users\Lenovo\networkprogramming2025\week07-store-forward-basic> python node.py
[NODE 8002] Listening...
[NODE 8002] Received: Hello from node 8001 from ('127.0.0.1', 57298)
[NODE 8002] Received: Hello from node 8000 from ('127.0.0.1', 49736)
```

Week 8

python node.py 8000 8001 8002

```
PS C:\Users\Lenovo\networkprogramming2025\week08-opportunistic-routing-basic> python node.py 8000 8001 8002
[NODE 8000] Candidates: []
[NODE 8000] Listening...
[NODE 8000] Store message for 8001
[NODE 8000] Store message for 8002
[NODE 8000] Candidates: [8001, 8002]
[NODE 8000] Received NEW: 8001:1775563917.9456372|Hello from node 8001
[NODE 8000] Sent → 8001
[NODE 8000] Forwarded message to 8001
[NODE 8000] Received NEW: 8000:1775563912.8632488|Hello from node 8000
[NODE 8000] Received NEW: 8001:1775563917.9536145|Hello from node 8001
[NODE 8000] Candidates: [8001, 8002]
[NODE 8000] Sent → 8001
[NODE 8000] Forwarded message to 8001
[NODE 8000] Sent → 8001
[NODE 8000] Forwarded message to 8001
[NODE 8000] Sent → 8001
[NODE 8000] Forwarded message to 8001
[NODE 8000] Sent → 8001
[NODE 8000] Forwarded message to 8001
[NODE 8000] Received NEW: 8002:1775563925.9550626|Hello from node 8002
[NODE 8000] Received NEW: 8000:1775563910.8511713|Hello from node 8000
[NODE 8000] Duplicate ignored
[NODE 8000] Duplicate ignored
[NODE 8000] Received NEW: 8002:1775563925.9558997|Hello from node 8002
[NODE 8000] Candidates: [8001, 8002]
[NODE 8000] Sent → 8001
[NODE 8000] Forwarded message to 8001
[NODE 8000] Sent → 8001
[NODE 8000] Forwarded message to 8001
[NODE 8000] Sent → 8001
[NODE 8000] Forwarded message to 8001
[NODE 8000] Duplicate ignored
[NODE 8000] Candidates: [8001, 8002]
```

python node.py 8001 8000 8002

```
PS C:\Users\Lenovo\networkprogramming2025\week08-opportunistic-routing-basic> python node.py 8001 8000 8002
[NODE 8001] Candidates: []
[NODE 8001] Listening...
[NODE 8001] Sent → 8000
[NODE 8001] Received NEW: 8000:1775563912.8632488|Hello from node 8000
[NODE 8001] Store message for 8002
[NODE 8001] Candidates: [8000, 8002]
[NODE 8001] Sent → 8000
[NODE 8001] Forwarded message to 8000
[NODE 8001] Sent → 8000
[NODE 8001] Forwarded message to 8000
[NODE 8001] Received NEW: 8000:1775563910.8511713|Hello from node 8000
[NODE 8001] Received NEW: 8001:1775563917.9456372|Hello from node 8001
[NODE 8001] Duplicate ignored
[NODE 8001] Received NEW: 8001:1775563917.9536145|Hello from node 8001
[NODE 8001] Received NEW: 8002:1775563925.9558997|Hello from node 8002
[NODE 8001] Candidates: [8000, 8002]
[NODE 8001] Sent → 8000
[NODE 8001] Forwarded message to 8000
[NODE 8001] Sent → 8000
[NODE 8001] Forwarded message to 8000
[NODE 8001] Sent → 8000
[NODE 8001] Forwarded message to 8000
[NODE 8001] Sent → 8000
[NODE 8001] Forwarded message to 8000
[NODE 8001] Received NEW: 8002:1775563925.9550626|Hello from node 8002
[NODE 8001] Duplicate ignored
[NODE 8001] Duplicate ignored
[NODE 8001] Candidates: [8000, 8002]
[NODE 8001] Sent → 8000
[NODE 8001] Forwarded message to 8000
[NODE 8001] Candidates: [8000, 8002]
```

python node.py 8002 8000 8001

```
PS C:\Users\Lenovo\networkprogramming2025\week08-opportunistic-routing-basic> python node.py 8002 8000 8001
[NODE 8002] Candidates: []
[NODE 8002] Listening...
[NODE 8002] Sent → 8000
[NODE 8002] Sent → 8001
[NODE 8002] Candidates: [8000, 8001]
```

Week 9

python node.py 7000 7001 7002

```
PS C:\Users\Lenovo\networkprogramming2025\week09-bio-routing-basic> python node.py 7000 7001 7002
[PHEROMONE TABLE] {}
[NODE 7000] Candidates: [7001, 7002]
[NODE 7000] Listening...
[NODE 7000] Failed to send to 7001
[NODE 7000] Failed to send to 7002
[PHEROMONE TABLE] {7001: 0.9, 7002: 0.9}
[NODE 7000] Candidates: [7001, 7002]
[NODE 7000] Received: Hello from node 7001 -> via 7001
[NODE 7000] Sent: Hello from node 7000 -> via 7000 to 7001
[NODE 7000] Sent: Hello from node 7000 -> via 7000 to 7001
[NODE 7000] Received: Hello from node 7000 -> via 7000 -> via 7001
[NODE 7000] Received: Hello from node 7000 -> via 7000 -> via 7001
[NODE 7000] Received: Hello from node 7001 -> via 7001
[PHEROMONE TABLE] {7001: 1.17, 7002: 0.81}
[NODE 7000] Candidates: [7001, 7002]
[NODE 7000] Sent: Hello from node 7001 -> via 7001 -> via 7000 to 7001
[NODE 7000] Sent: Hello from node 7001 -> via 7001 -> via 7000 to 7001
[NODE 7000] Received: Hello from node 7002 -> via 7002
[NODE 7000] Received: Hello from node 7002 -> via 7002 -> via 7001
[PHEROMONE TABLE] {7001: 1.41, 7002: 0.73}
[NODE 7000] Candidates: [7001, 7002]
[NODE 7000] Sent: Hello from node 7002 -> via 7002 -> via 7000 to 7001
[NODE 7000] Sent: Hello from node 7002 -> via 7002 -> via 7001 -> via 7000 to 7001
[NODE 7000] Received: Hello from node 7002 -> via 7002 -> via 7000 -> via 7001
[PHEROMONE TABLE] {7001: 1.63, 7002: 0.66}
[NODE 7000] Candidates: [7001, 7002]
[PHEROMONE TABLE] {7001: 1.47, 7002: 0.59}
[NODE 7000] Candidates: [7001, 7002]
```

python node.py 7001 7000 7002

```
PS C:\Users\Lenovo\networkprogramming2025\week09-bio-routing-basic> python node.py 7001 7000 7002
[PHEROMONE TABLE] {}
[NODE 7001] Listening...
[NODE 7001] Candidates: [7000, 7002]
[NODE 7001] Sent: Hello from node 7001 -> via 7001 to 7000
[NODE 7001] Received: Hello from node 7000 -> via 7000
[NODE 7001] Received: Hello from node 7000 -> via 7000
[NODE 7001] Failed to send to 7002
[PHEROMONE TABLE] {7000: 1.08, 7002: 0.9}
[NODE 7001] Candidates: [7000, 7002]
[NODE 7001] Sent: Hello from node 7000 -> via 7000 -> via 7001 to 7000
[NODE 7001] Sent: Hello from node 7000 -> via 7000 -> via 7001 to 7000
[NODE 7001] Sent: Hello from node 7001 -> via 7001 to 7000
[NODE 7001] Received: Hello from node 7001 -> via 7001 -> via 7000
[NODE 7001] Received: Hello from node 7001 -> via 7001 -> via 7000
[NODE 7001] Received: Hello from node 7002 -> via 7002
[PHEROMONE TABLE] {7000: 1.51, 7002: 0.81}
[NODE 7001] Candidates: [7000, 7002]
[NODE 7001] Sent: Hello from node 7002 -> via 7002 -> via 7001 to 7000
[NODE 7001] Received: Hello from node 7002 -> via 7002 -> via 7000
[NODE 7001] Received: Hello from node 7002 -> via 7002 -> via 7001 -> via 7000
[PHEROMONE TABLE] {7000: 1.54, 7002: 0.73}
[NODE 7001] Candidates: [7000, 7002]
[NODE 7001] Sent: Hello from node 7002 -> via 7002 -> via 7000 -> via 7001 to 7000
[PHEROMONE TABLE] {7000: 1.57, 7002: 0.66}
[NODE 7001] Candidates: [7000, 7002]
[PHEROMONE TABLE] {7000: 1.41, 7002: 0.59}
[NODE 7001] Candidates: [7000, 7002]
[PHEROMONE TABLE] {7000: 1.27, 7002: 0.53}
[NODE 7001] Candidates: [7000, 7002]
```

python node.py 7002 7000 7001

```
PS C:\Users\Lenovo\networkprogramming2025\week09-bio-routing-basic> python node.py 7002 7000 7001
[PHEROMONE TABLE] {}
[NODE 7002] Candidates: [7000, 7001]
[NODE 7002] Listening...
[NODE 7002] Sent: Hello from node 7002 -> via 7002 to 7000
[NODE 7002] Sent: Hello from node 7002 -> via 7002 to 7001
[PHEROMONE TABLE] {7000: 1.08, 7001: 1.08}
[NODE 7002] Candidates: [7000, 7001]
[PHEROMONE TABLE] {7000: 0.97, 7001: 0.97}
[NODE 7002] Candidates: [7000, 7001]
[PHEROMONE TABLE] {7000: 0.87, 7001: 0.87}
[NODE 7002] Candidates: [7000, 7001]
[PHEROMONE TABLE] {7000: 0.79, 7001: 0.79}
[NODE 7002] Candidates: [7000, 7001]
[PHEROMONE TABLE] {7000: 0.71, 7001: 0.71}
[NODE 7002] Candidates: [7000, 7001]
[PHEROMONE TABLE] {7000: 0.64, 7001: 0.64}
[NODE 7002] Candidates: [7000, 7001]
[PHEROMONE TABLE] {7000: 0.57, 7001: 0.57}
[NODE 7002] Candidates: [7000, 7001]
```

Week 10

python node.py 8000 8001 8002

```
PS C:\Users\Lenovo\networkprogramming2025\week10-quantum-network-basic> python node.py 8000 8001 8002
[NODE 8000] Listening...
[NODE 8000] Sent token to 8001
[NODE 8000] Received & consumed: Quantum token from 8001 -> via 8001
[NODE 8000] Received & consumed: Quantum token from 8000 -> via 8000 -> via 8001
[NODE 8000] Sent token to 8001
[NODE 8000] Received & consumed: Quantum token from 8002 -> via 8002
[NODE 8000] Sent token to 8001
[NODE 8000] Received & consumed: Quantum token from 8002 -> via 8002 -> via 8000 -> via 8001
[NODE 8000] Dropped expired token
```

python node.py 8001 8000 8002

```
PS C:\Users\Lenovo\networkprogramming2025\week10-quantum-network-basic> python node.py 8001 8000 8002
[NODE 8001] Listening...
[NODE 8001] Received & consumed: Quantum token from 8000 -> via 8000
[NODE 8001] Sent token to 8000
[NODE 8001] Sent token to 8000
[NODE 8001] Received & consumed: Quantum token from 8001 -> via 8001 -> via 8000
[NODE 8001] Received & consumed: Quantum token from 8002 -> via 8002 -> via 8000
[NODE 8001] Sent token to 8000
[NODE 8001] Dropped expired token
```

python node.py 8002 8000 8001

```
PS C:\Users\Lenovo\networkprogramming2025\week10-quantum-network-basic> python node.py 8002 8000 8001
[NODE 8002] Listening...
[NODE 8002] Sent token to 8000
```